

Lab 3 Calculations

3.A RMS Voltage calculation

3.A.1 Sine wave

$$\begin{aligned}V_{\text{magnitude}} &= V_{\text{peak-to-peak}} \times \frac{1}{2} = V_{\text{RMS}} \times \sqrt{2} \\10V \times \frac{1}{2} &= V_{\text{RMS}} \times \sqrt{2} \\V_{\text{RMS}} &= \frac{5}{\sqrt{2}} V \approx\end{aligned}$$

3.A.2 Square wave

$$\begin{aligned}V_{\text{magnitude}} &= V_{\text{peak-to-peak}} \times \frac{1}{2} = V_{\text{RMS}} \\10V \times \frac{1}{2} &= V_{\text{RMS}} \\V_{\text{RMS}} &= 5V \approx\end{aligned}$$

3.A.3 Triangle wave

$$\begin{aligned}V_{\text{magnitude}} &= V_{\text{peak-to-peak}} \times \frac{1}{2} = V_{\text{RMS}} \times \sqrt{3} \\10V \times \frac{1}{2} &= V_{\text{RMS}} \times \sqrt{3} \\V_{\text{RMS}} &= \frac{5}{\sqrt{3}} V \approx\end{aligned}$$

3.B Period calculation

3.B.1 100 Hz

$$\begin{aligned}\text{frequency} &= \frac{1}{\text{period}} \\100 \text{ Hz} &= \frac{1}{\text{period}} \\\text{period} &= \frac{1}{100 \text{ Hz}} = 0.01s = 10ms\end{aligned}$$

3.B.2 1 kHz

$$\begin{aligned}\text{frequency} &= \frac{1}{\text{period}} \\ 1 \text{ kHz} = 1000 \text{ Hz} &= \frac{1}{\text{period}} \\ \text{period} &= \frac{1}{1000 \text{ Hz}} = 0.001s = 1ms\end{aligned}$$

3.B.3 100 kHz

$$\begin{aligned}\text{frequency} &= \frac{1}{\text{period}} \\ 100 \text{ kHz} = 100000 \text{ Hz} &= \frac{1}{\text{period}} \\ \text{period} &= \frac{1}{100000 \text{ Hz}} = 0.00001s = 10\mu s\end{aligned}$$